

NMFAD: Neighbor-Aware Mask-Filling Attributed Network Anomaly Detection

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Abstract—As a widely adopted protocol for anomaly detection in attributed networks, reconstruction error prioritizes comprehensive feature extraction to detect anomalies over interrogating the differential representation between normal and abnormal nodes. Intuitively, in attributed networks, normal nodes and their neighbors often exhibit similarities, whereas abnormal nodes demonstrate behaviors distinct from their neighbors. Hence, normal nodes can be accurately represented through their neighbors and effectively reconstructed. As opposed to normal nodes, abnormal nodes represented by their neighbors may be erroneously reconstructed as normal, resulting in increased reconstruction error. Leveraging from this observation, we propose a novel anomaly detection protocol called Neighbor-aware Mask-Filling Anomaly Detection (NMFAD) for attributed networks, aiming to maximize the variability between original and reconstructed features of abnormal nodes filled with information from their neighbors. Specifically, we utilize random-mask on nodes and integrate them into the backbone Graph Neural Networks (GNNs) to map nodes into a latent space. Subsequently, we fill the masked nodes with embeddings from their neighbors and smooth the abnormal nodes closer to the distribution of normal nodes. This optimization improves the likelihood of the decoder to reconstructing abnormal nodes as normal, thereby maximizing the reconstruction error of abnormal nodes. Experimental results demonstrate that, compared to the existing models, NMFAD exhibits superior performance in attributed networks.

Index Terms—Anomaly detection, attributed network, graph convolutional network, reconstruction, mask-filling, neighborhood association.

I. INTRODUCTION

ATTRIBUTED network refers to a network in which the nodes or edges contain attribute information. It is a typical type of graph data where nodes represent entities and edges represent relationships between entities [1], [2], [3]. Anomaly detection in attributed networks aims to

identify abnormal nodes, which has been widely used in many fields, such as financial fraud [4], intrusion detection, spam detection [5], and others [6]. Unlike time series data, tabular data, etc, anomaly detection in graph data needs to confront more challenges due to the particularity of graph structure, indicating the complex “many-to-many” relationship between nodes, and usually contain more information.

The traditional attributed network anomaly detection methods [7], [8] utilize shallow mechanisms to capture the network structural information and node attributes. However, those methods are unable to be applied in the case of the increasing size of the data. Recently, Graph Neural Networks (GNNs) [9], [10], [11], [12] have been proposed to handle the graph data. There are mainly 3 categories of graph anomaly detection methods, one-class [13], [14], self-supervised [15], [16], [17], [18], and reconstruction error [19], [20]. They all have achieved promising results on attributed network anomaly detection, but most of them have some limitations in the actual attributed networks: (1) Most methods mine the attributed network information by designing complex network structures for inter-node interaction learning [20], [21] rather than directly designing appropriate anomaly detection protocols. (2) A direct consequence is that, after sufficient representation learning, the anomalies might be also reconstructed well, resulting in the failure of the anomaly score based on reconstruction error [1].

Apparently, in attributed networks, normal nodes and their neighbors tend to be similar, while abnormal nodes behave differently from their neighbors. That is, normal nodes and their neighbors tend to have similar distribution states in terms of structure and attributes. Therefore, we can assert that using the neighbor information for filling of masked nodes could effectively make the reconstructed abnormal nodes inconsistent with their original ones, while the reconstructed normal nodes similar to their original ones, thus, maximizing the difference between normal and abnormal nodes in reconstruction error, which can be illustrated in Fig. 1. In light of this motivation, we propose a novel Neighbor-aware Mask-Filling Attributed Network Anomaly Detection method (NMFAD). Specifically, we perform random mask of nodes and input them into the encoder to extract the embeddings. Then, we fill the masked nodes with neighbor embedding information and smooth the distribution using Kullback-Leibler (KL) divergence regularization to make the masked abnormal nodes closer to the distribution of normal nodes. Finally, we process the attributed network anomaly detection tasks using reconstruction error.

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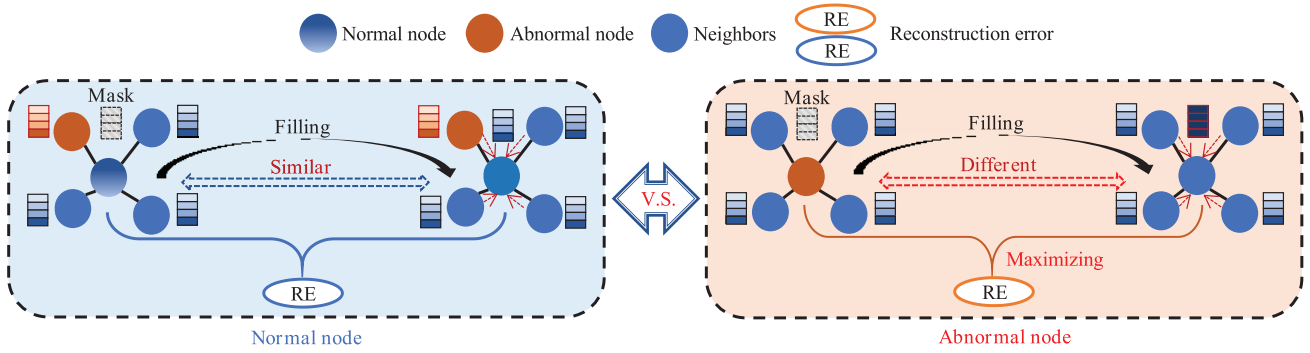


Fig. 1. The motivation of NMFAD.

The main contributions of this paper include the following:

- 1) We propose a novel anomaly detection protocol, by maximizing the variability of abnormal node between its original features and reconstructed features filled from neighbors, due to the dissimilarity between abnormal nodes and neighbors.
- 2) We design the strategy of mask-filling with distribution smoothing to draw the mask-filled abnormal nodes closer to the distribution of normal nodes.
- 3) We conducted different experiments on four datasets of two attributed network anomaly detection scenarios to validate the advantages of NMFAD over baseline methods.

II. RELATED WORK

A. Traditional Anomaly Detection

Presently, the majority of anomaly detection methods are unsupervised and aim to identify anomalies when confronted with a substantial amount of unlabeled data alongside limited labeled anomaly data [22]. These methods can be categorized into 3 groups: clustering, one-class, and reconstruction.

- 1) Clustering-based methods typically follow a two-step approach [23], [24], involving dimensionality reduction followed by anomaly detection through clustering of normal samples.
- 2) In contrast, one-class-based methods [25], [26], [27], [28] focus on learning the normal/abnormal discriminative boundary. One-Class SVM (OC-SVM) [25] seeks to fit a hyperplane in feature space transformed non-linearly to enclose normal data. Support Vector Data Description (SVDD) learns a hypersphere around normal data to enhance the robustness. Recently, deep learning-based methods that aim to capture the embeddings and hyperspheres have been introduced. A typical approach is Deep-SVDD [26].
- 3) Reconstruction-based methods operate under the assumption that abnormal samples cannot be effectively reconstructed. This category includes PCA (Principal Component Analysis)-based [29] and AE (AutoEncoder)-based methods [30] which leverage the reconstruction error as an anomaly score.

While these methods have demonstrated success in traditional anomaly detection, they lack generalization ability for

graph data, where the topology structure among nodes plays a vital role [13], [31].

B. Attributed Network Embeddings

Attributed networks are prevalent in the real world, and various methods focus on learning the node features by incorporating both topological structure and node attributes [32] for downstream tasks such as anomaly detection. Existing approaches mainly include 3 strategies: random walk [8], matrix factorization, and deep learning.

- 1) Among the random-walk-based methods, SANE (Sparse Attributed Network Embedding) [33] generates node sequences through the random walk and aggregates the information from neighbors with attention mechanism, to learn the low-dimensional features. Text-Associated DeepWalk (TADW) [34] integrates text features to improve DeepWalk. Extending TADW, HSCA (Homophily, Structure, and Content Augmented) [35] introduces regularization of neighboring nodes to capture network homogeneity.
- 2) Matrix-factorization-based methods, like AANE (Accelerated Attributed Network Embedding) [36] and BANE (Binarized Attributed Network Embedding) [37], combine node attribute and edge information from neighboring nodes to capture the association between structure and attribute information.
- 3) Deep-learning-based methods [38], [39] aim to learn node embeddings and show excellent performance. HNE (Heterogeneous Network Embedding) [40] maps the nodes and attributes into a shared latent space for graph learning. CSAN (Co-embedding for Static Attributed Networks) [41] adopts a joint learning framework with Variational AutoEncoder (VAE) to extract embeddings.

While these methods have shown promising performance in attributed network embeddings, they still focus on feature extraction without a targeted design for graph anomaly detection tasks.

C. Anomaly Detection in Attributed Network

The advent of deep learning and its application to graph data scenarios has revolutionized graph anomaly detection [42]. Graph Neural Networks (GNNs), such as GCN, GAT (Graph Attention Network), etc [9], [10], [11], [12], have performed superior performance in various graph mining tasks recently.

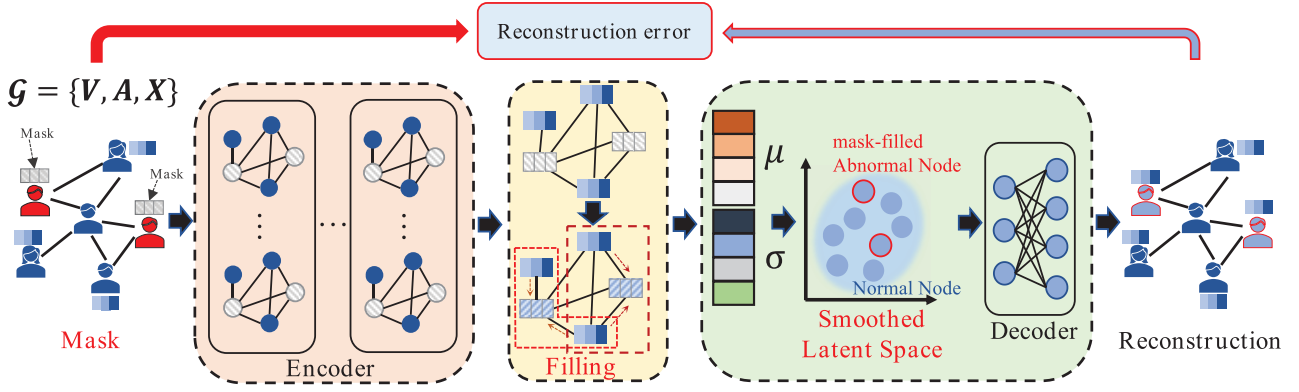


Fig. 2. The framework of NMFAD.

GNNs can learn node embeddings in a graph, with both node attributes and the graph structure [13], [17]. Currently, the related methods include 3 main categories: one-class, self-supervised, and reconstruction error.

- 1) The one-class graph anomaly detection methods such as OC-GNN [16] and Dual-SVDAE [12], combine the advantage of GNNs for fully capturing graph information and the one-class protocol for anomaly detection by a learned hypersphere.
- 2) The self-supervised methods such as GraphMAE [16] and MGAE [43], utilize the prior knowledge or data augmentation mechanisms to learn the graph information and capture the normal/abnormal difference by a designed anomaly score.
- 3) Besides, the reconstruction-error-based methods such as Dominant [19] and AnomalyDAE [20], assume that the abnormal nodes have larger reconstruction errors compared with normal nodes after fully capturing the data features. These methods mainly rely on the AE frameworks, which are simple and effective, but they pay more attention to full feature extraction for anomaly detection rather than discovering the normal/abnormal difference. the graph data mining rather than the difference information between normal and abnormal capturing.

Therefore, we explore the normal/abnormal variability from the perspective of neighbor perception in the attributed networks, consider the similarity of normal nodes and neighbors, and the dissimilarity of abnormal nodes and their neighbors, combine the advantage of reconstruction error for anomaly detection, This is the core idea of our approach.

III. METHODOLOGY

The framework of NMFAD is an AutoEncoder structure, as shown in Fig. 2, and the main parameters are listed in Table I. Firstly, we treat the nodes and edges separately, and obtain the node embeddings by Graph AutoEncoder (GAE) with random mask. Then, we select the neighbor embeddings of each masked node to fill it and smooth the distribution of mask-filled abnormal nodes with normal ones based on KL divergence, making the reconstructed abnormal nodes more like normal rather than their original abnormal nodes. Finally,

TABLE I
MAIN PARAMETER LIST

Parameter	Description
$\mathbf{G}$	Graph of attributed network.
V	Node set in attributed network.
$\mathbf{A}$	Adjacency matrix of attributed network.
$\mathbf{X}$	Node feature matrix.
$\mathbf{H}$	Node embedding matrix.
N_i	Neighbor number of the i th node.
$\mathbf{G}'$	Reconstructed graph of the attributed network, $\mathbf{G}$.
$\mathbf{x}_i$	Original node features.
$\mathbf{x}_i^{[M]}$	Masked node features of $\mathbf{x}_i$.
$\mathbf{x}'_i$	Reconstructed node features of $\mathbf{x}_i$.

we use the reconstruction error with the Scaled Cosine Error (SCE) for attributed network anomaly detection.

A. Node Embeddings by GAE With Random Mask

Given a graph, $\mathbf{G} = (\mathbf{V}, \mathbf{A}, \mathbf{X})$, which represents the attributed network, where V denotes the node set, $\mathbf{A} \in \{0, 1\}^{N \times N}$ is the adjacency matrix, $\mathbf{X} \in \mathbb{R}^{N \times d}$ is the node feature matrix, N is the node number, and d is the attribute dimension.

Firstly, using a random sampling strategy that follows a uniform distribution to prevent potential center bias, we mask the sampled nodes [44]. Given a node, $v_i \in V$, its feature vector, $\mathbf{x}_i$, could be defined as:

$$\tilde{\mathbf{x}}_i = \begin{cases} \mathbf{x}_i^{[M]}, & \text{for } v_i \in \tilde{V} \\ \mathbf{x}_i, & \text{for } v_i \notin \tilde{V} \end{cases} \quad (1)$$

where $\tilde{V} = \{\tilde{v}_1, \tilde{v}_2, \dots, \tilde{v}_{|\tilde{V}|}\}$ represents the masked node subset of V , $\mathbf{x}_i^{[M]} \in \mathbb{R}^d$ represents the masked features of $\mathbf{x}_i$, which is a learnable vector during model training. The objective function is used to reconstruct the masked node features.

In most cases, a relatively high mask ratio, β , can reduce redundancy in the graph structure and prevent the model from inferring the masked nodes based on the visible neighbors. Moreover, this strategy also can sparse the input to the encoder, as the encoder only processes visible nodes, reducing the computational requirements to train large-scale encoders at a lower cost [16].

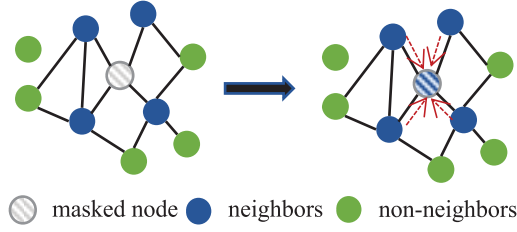


Fig. 3. Mask-filling with neighbors.

Furthermore, we adopt GAE to extract node embeddings with the following definitions: f_E is the graph encoder, f_D is the graph decoder, $H \in \mathbb{R}^{N \times d_h}$ denotes the node embedding matrix in latent space, d_h is the embedding dimension of latent space. Following MGAE [42], the encoder and decoder are constructed using GAT.

The graph encoder employs a graph attention (GAT) layer to aggregate inter-node association features, implementing shared attention at each node:

$$w_{i,j} = \text{attn}(\mathbf{x}_i, \mathbf{x}_j) = \sigma(\mathbf{a}^T \cdot [\mathbf{W}\mathbf{x}_i \parallel \mathbf{W}\mathbf{x}_j]) \quad (2)$$

where $w_{i,j}$ denotes the weight, $\text{attn}(\cdot)$ represents the shared weight of nodes parameterized by $\mathbf{a} \in \mathbb{R}^d$ and $\mathbf{W} = \mathbb{R}^{\frac{d}{2} \times F}$, where d is the neuron number, F is the embedding dimension, and $\parallel$ represents the concatenation operation. $\sigma(\cdot)$ is the activation function. The final node attention weight, $\alpha_{i,j}$, is normalized using $\text{softmax}(\cdot)$:

$$\alpha_{i,j} = \frac{e^{w_{i,j}}}{\sum_{k \in N_i} e^{w_{i,k}}} \quad (3)$$

where N_i denotes the number of neighbors of v_i , which is obtained from $\mathbf{A}$.

Finally, the learned $\alpha_{i,j}$ is combined with $\mathbf{X}$ to produce the embedding, $\mathbf{H}_i$ of v_i :

$$\mathbf{H}_i = \sum_{k \in N_i} \alpha_{i,k} \cdot \mathbf{x}_k. \quad (4)$$

So, we can extract the node embedding matrix, $\mathbf{H} = f_E(\mathbf{A}, \mathbf{X})$, to reconstruct the graph of the attributed network by decoder, $\mathbf{G}' = f_D(\mathbf{A}, \mathbf{H})$, which has a similar structure to the encoder, to obtain the reconstructed nodes, $\mathbf{X}' = \{\mathbf{x}'_1, \mathbf{x}'_2, \dots, \mathbf{x}'_N\}$.

B. Mask-Filling and Distribution Smoothing

The motivation of NMFAD is the reconstruction effects of mask-filled abnormal features are significantly different from their original ones. Therefore, before decoding, we design mask-filling and distribution smoothing strategies to implement this idea and drive the decoder to reconstruct abnormal nodes as normal rather than their original ones to maximize the reconstruction error of abnormal nodes.

Firstly, we fill the masked node embeddings with the mean value of the neighbors' embeddings, as demonstrated in Fig. 3.

$$\mathbf{x}_i^{[M]} = \frac{1}{N_i} \sum_{j \in N_i} \mathbf{x}_i^j \quad (5)$$

where $\mathbf{x}_i^j \in \mathbf{X}_i^j$, $\mathbf{X}_i^j$ is the neighbor embedding matrix of $\tilde{v}_i$. Secondly, we calculate the mean and variance of the filled embeddings through a fully connected layer, and adopt KL divergence to smooth the abnormal nodes closer to the distribution of normal nodes:

$$\mathcal{L}_{KL} = -KL(q(\mathbf{H}|\mathbf{X}, \mathbf{A}) \| p(\mathbf{H})) \quad (6)$$

where $q(\mathbf{H}|\mathbf{X}, \mathbf{A})$ is the learned node distribution, and $p(\mathbf{H})$ is the prior distribution, which is selected as the Gaussian distribution in this paper.

C. Anomaly Detection in Attributed Network

The motivation of NMFAD is based on the variability between original features and reconstructed features of abnormal nodes in the attributed network, so the reconstruction error is chosen as the anomaly score. Given a node, v_i , the anomaly score, $S(v_i)$, is defined as follows:

$$S(v_i) = \|\mathbf{x}_i - \mathbf{x}'_i\| \quad (7)$$

The node with a small reconstruction error is detected as normal, otherwise, it is detected as abnormal.

The traditional method for calculating anomaly scores is Mean Square Error (MSE). However, MSE could be minimized to nearly 0, and it has the following problems: (1) Sensitivity to vector norm and dimensionality, and (2) Low selectivity to focus on those harder samples among imbalanced easy-and-hard (i.e., easily reconstructed or not) samples. So we follow GraphMAE (Masked Graph AutoEncoder) and use Scaled Cosine Error (SCE) for reconstruction error calculation [12], which could reduce the weights of easier samples by scaling the cosine error with $\gamma \geq 1$. In particular, MSE is SCE when $\gamma = 1$.

SCE is defined as follows:

$$\mathcal{L}_{SCE} = \frac{1}{|\tilde{V}|} \sum_{v_i \in \tilde{V}} \left(1 - \frac{\mathbf{x}_i^T \mathbf{x}'_i}{\|\mathbf{x}_i\| \cdot \|\mathbf{x}'_i\|} \right)^\gamma, \gamma \geq 1 \quad (8)$$

where γ is the scaling factor.

In summary, the loss function of NMFAD is:

$$\mathcal{L} = \mathcal{L}_{SCE} + \mathcal{L}_{KL}. \quad (9)$$

The training process of NMFAD is shown as the following Algorithm.

IV. EXPERIMENT

A. Experimental Setup

The experiment are built in PyTorch and run on an Ubuntu (18.04) OS Server:

- CPU: Intel Xeon Gold 5320
- GPU: NVIDIA A40 Graphics card
- Memory: 256G RAM

We choose four attributed network datasets (Cora and Citeseer are citation networks, and Flickr and BlogCatalog are social networks) for our experiments. The main statistics are listed in Table II. Following the experimental protocol in [1] [13], and [14], on each dataset, we choose one class of nodes as the normal ones at random, and the others as the

Algorithm 1 NMFAD training process.**Require:** Attributed network, $G=(V,A,X)$;Weight sharing GAT encoder, f_E ;The number of iterations, T **BEGIN**

- 1: **for** iterator = 1, 2, 3, ..., T **do**
- 2: Randomly sample and mask nodes via Eq. (1);
- 3: Normalize the attention weight of nodes, $\alpha_{i,j}$, using $\text{softmax}(\cdot)$ via Eq. (3);
- 4: Obtain the embedding matrix, H , via Eq. (4);
- 5: Mask-fill the masked nodes via Eq. (5).
- 6: Smooth the distribution using KL divergence via Eq. (6);
- 7: Update NMFAD by optimizing Eq. (9).
- 8: **end for**
- 9: **return** the updated f_E

END

TABLE II
THE STATISTICS OF FOUR DATASETS

Dataset	Node	Edge	Attribute	Class
Cora	2708	5429	1433	7
Citeseer	3327	4732	3703	6
Flickr	7575	239738	12407	9
BlogCatalog	5196	121743	8189	6

abnormal nodes. For normal nodes, we randomly select 60% for training, 15% for validation, and 25% for testing. Then, in the validation and test sub-set, 50% of the nodes are normal, and the others are randomly selected from the abnormal nodes.

- 1) **Cora**¹: It consists of 2708 samples, each of which is a scientific publication. All the samples are divided into 7 categories. The total number of citation links (edges) is 5429. Each sample is represented by a 1433-dimensional word vector, corresponding to a lexicon containing 1433 words, and the value of each dimension (0 or 1) corresponds to whether the word appears in this paper or not.
- 2) **Citeseer**²: It has 4732 edges (citation links) among 3312 scientific publications (samples), which were divided into 6 categories. Each publication is represented in the same way as the Cora dataset, with a dimension of 3703 after removing the less frequent words.
- 3) **Flickr**³: It is an image hosting and sharing network. It consists of 7575 nodes and 239738 edges. Each node represents a user, and each edge represents a friend-relations. In addition, each user has a label of interest group (total: 9 groups). Node attributes are represented as the users generate personalized content.
- 4) **BlogCatalog**⁴: It is a Blog sharing network, consisting of 5196 nodes and 121743 edges. The nodes represent the bloggers, and the edges represent some kind of friendship. The blogger's label represents his/her interest

(total: 6 interests). The attributes of each node are generated in the same way as the Flickr dataset.

B. Comparative Models

We select some currently known deep learning-based methods to compare with NMFAD. Among them, OCSVM, Deep-SVDD, and DeepWalk are classical one-class methods; GAE, ARG, Dominant, and ComGA are reconstruction-error-based methods; OC-GNN and Dual-SVDAE are one-class GNN methods; CCA-SSG, GraphMAE, and CoLA are self-supervised methods.

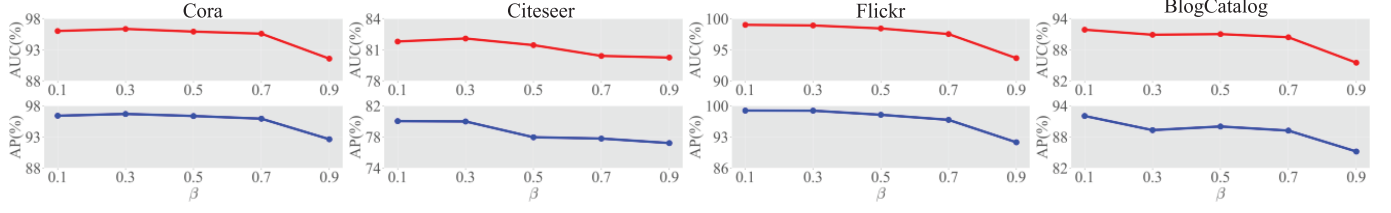
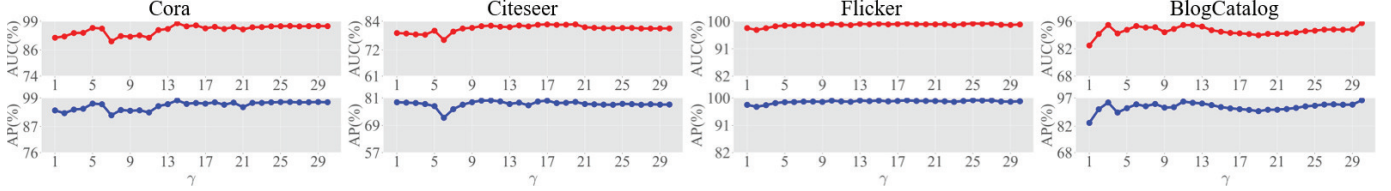
- 1) **OC-SVM** (2001) [25]: It is a classical anomaly detection algorithm with a hyperplane to detect the anomalies.
- 2) **Deep-SVDD** (2018) [26]: It is a deep one-class anomaly detection method with SVDD. It has two strategies, **Deep-SVDD (Attr)** and **Deep-SVDD (Stru)**, which use the attribute and structure information for model training, respectively.
- 3) **DeepWalk** (2014) [8]: It is a classical graph embedding method, using local information through truncated random walks to extract the embeddings and employ OC-SVM to process anomaly detection.
- 4) **GAE** (2020) [41]: It is a generative graph self-supervised method, which could capture the high non-linearity of topological structure and node attributes, and perform anomaly detection with the reconstruction error.
- 5) **ARGA** (Adversarially Regularized Graph Autoencoder, 2018) [38]: It is an adversarial graph embedding framework based on GAE to enforce the embeddings of topological structure and node attributes to match a prior distribution.
- 6) **Dominant** (2019) [19]: It combines GCN with deep AutoEncoder to reconstruct the attributed network through topological structure and node attributes, and spot the anomalies.
- 7) **OC-GNN** (2021) [13]: It is a GNN-based one-class graph anomaly detection framework, such as GCN (called **OC-GCN**) or GraphSage (called **OC-SAGE**).
- 8) **Dual-SVDAE** (2022) [14]: It contains two AEs to fuse the structure and attribute embeddings. The anomalous nodes can be measured by the learned structure and attribute hyperspheres, respectively.
- 9) **CoLA** (CONtrastive Learning for Anomaly detection, 2022) [17]: It is a Contrastive self-supervised Learning Anomaly detection framework that extracts the relations between nodes and their neighboring substructures.
- 10) **CCA-SSG** (Canonical Correlation Analysis to Self-Supervised GNN, 2021) [15]: It is a self-supervised graph embedding learning model, which uses canonical correlation analysis to associate two view embeddings through data augmentation. It also could discard the augmentation-variant information and prevent degenerated solutions.
- 11) **GraphMAE** (2022) [16]: It is a GAE with a mask strategy, focusing on feature reconstruction with SCE.

¹<https://linqs.org/datasets/#cora>

²<https://linqs.org/datasets/#citeseer-doc-classification>

³<http://shannon.cs.illinois.edu/DenotationGraph>

⁴<http://socialcomputing.asu.edu/datasets/BlogCatalog>


 Fig. 4. The experimental results of NMFAD with different values of β on different datasets.

 Fig. 5. The experimental results of NMFAD with different values of γ on different datasets.

- 12) **CONAD**(CONtrastive Anomaly Detection, 2022) [45]: It is a contrastive learning framework for attributed network anomaly detection with human knowledge.
- 13) **ComGA** (2022) [46]: It is a new community-aware attribution network anomaly detection framework that designs a tailored deep GCN (tGCN) to detect anomalies.

C. Model Settings and Performance Metrics

In NMFAD, the learning rate is set to 0.01, and the epoch is set to 200. On each dataset, we set the dimension of embeddings as 64. We use Adam Optimizer for training.

We adopt two performance metrics to evaluate the performance: AUC and AP. All the experimental results were obtained by averaging the results with five different seeds.

D. Parameter Experiments

We first discuss the effects of different mask ratio, β , and set the value of the scaling factor of SCE, γ , in the loss function. The experimental results are shown in Fig. 4 and Fig. 5.

- 1) In Fig. 4, the experimental results gradually decrease as β increases on each dataset, but the overall trend is stable. This shows that even when β is set to a larger value, we can lend the information filling of neighbors to ensure the NMFAD's ability to reconstruct the masked normal samples. We set β to 0.3 on each dataset for the subsequent experiments.
- 2) In Fig. 5, as γ increases, the detection performance gradually stabilizes from a certain initial fluctuation on Cora, Citeseer and BlogCatalog datasets. On Flickr dataset, The detection performance smooths out rapidly as γ increases. These results also could reflect that γ needs to be set to a larger value to reduce the weights of easier samples for enhance the detection performance. Moreover, this is why we chose SCE instead of MSE as the reconstruction error calculation. In the subsequent experiments, referring to the results in stable phase, we

set γ to 14 on Cora, Flickr and Citeseer datasets, and set γ to 12 on BlogCatalog dataset, respectively.

E. Comparative Experiment

The experimental results are shown in Table III. We can see that NMFAD overall outperforms other methods.

- 1) Some of the methods perform poorly on each dataset, such as OC-SVM, Deep-SVDD, DeepWalk, GAE and CoLA; some of the methods perform well on some datasets but poorly on others, such as ARGa and Dominant on Flickr dataset, OC-GCN and CCA-SSG on Citeseer dataset; while some of the methods perform better on each dataset, but not as well as NMFAD, such as OC-SAGE, Dual-SVDAE, and GraphMAE.
- 2) The mean results of ComGA are better than NMFAD's on Citeseer dataset, but in terms of standard deviation, ComGA is very unstable (AUC: 5.84% vs. 1.02%; AP: 5.69% vs. 1.26%), and weaker than NMFAD on the other datasets. Moreover, NMFAD gets the second-best mean results on Citeseer dataset, significantly outperforming ComGA on Flickr (AUC: 98.91% vs. 57.88%; AP: 98.88% vs. 58.32%) and BlogCatalog (AUC: 91.99% vs. 68.25%; AP: 91.94% vs. 68.15%) datasets. In particular, ComGA is community-based and is more complex. Therefore, we can conclude that NMFAD outperforms than ComGA overall.
- 3) On each dataset, the standard deviation results of NMFAD are better than other methods, which can fully illustrate that NMFAD has better stability in different types of data scenarios.

In all, the normal/abnormal boundary of the one-class methods cannot effectively detect all abnormal nodes; the self-supervised methods require sufficient data augmentation which can not cover all anomalies; the reconstruction-error-based methods mainly improve the model ability from the feature extraction perspectives, without targeted anomaly detection protocol design, which can not be fully competent in the tasks of node anomaly detection in attribute networks. In

TABLE III
DETECTION RESULTS OF THESE METHODS. THE BEST RESULTS ARE MARKED IN BOLD, THE SECOND BEST RESULTS ARE ITALIC

Methods	Cora		Citeseer		Flickr		BlogCatalog	
	AUC	AP	AUC	AP	AUC	AP	AUC	AP
OC-SVM	63.30±5.96	59.91±5.05	51.04±1.52	52.17±0.95	32.52±1.47	38.78±0.59	56.21±1.99	55.81±1.73
Deep-SVDD(Attr)	67.49±7.34	66.04±6.01	54.82±7.19	53.62±6.78	85.36±6.24	78.34±8.43	63.70±2.97	61.39±3.33
Deep-SVDD(Stru)	69.55±4.20	65.80±5.34	49.95±1.68	51.39±2.32	85.47±3.48	85.57±4.60	67.56±4.45	63.95±2.41
DeepWalk	62.88±3.40	60.20±3.61	51.62±3.11	51.48±2.41	44.36±2.55	50.21±2.73	67.83±3.05	63.67±3.33
GAE	60.58±4.64	60.95±4.60	47.87±4.18	48.74±3.26	35.70±3.25	42.58±2.16	54.39±2.88	55.04±2.79
ARGA	93.41±0.56	90.76±3.02	81.12±1.89	80.42±1.68	62.13±2.87	65.74±4.61	80.23±3.23	80.01±3.42
Dominant	<i>94.86±3.26</i>	<i>95.64±2.09</i>	62.64±12.46	60.77±12.96	32.12±1.57	38.66±0.62	57.80±1.93	57.07±1.65
OC-GCN	70.03±4.83	66.80±5.20	49.91±4.26	48.99±3.87	83.80±5.41	82.53±5.92	64.18±5.65	60.78±4.15
OC-SAGE	83.74±1.92	81.83±2.97	61.63±1.94	62.20±2.61	81.28±4.59	78.93±4.80	67.32±3.67	66.75±2.85
Dual-SVDAE	83.50±4.22	84.96±4.77	67.88±4.06	66.96±5.34	88.06±2.43	84.57±3.56	<i>89.40±5.85</i>	<i>88.93±8.22</i>
CoLA	43.45±2.93	47.53±2.64	50.60±2.96	50.46±3.11	51.43±2.77	51.10±2.50	47.84±2.90	49.33±3.07
CCA-SSG	79.91±6.58	69.11±7.31	60.29±6.37	55.42±3.10	87.53±0.82	77.66±1.65	74.66±3.20	62.42±3.47
GraphMAE	86.21±5.50	86.60±5.30	74.68±2.68	73.61±1.71	<i>95.57±0.83</i>	<i>95.83±0.66</i>	65.87±2.97	66.37±3.36
CONAD	52.63±2.44	53.73±2.82	52.80±3.82	54.42±4.61	52.62±9.40	55.10±9.05	56.41±8.39	58.89±9.49
ComGA	93.60±2.39	92.25±2.72	85.69±5.84	84.64±5.69	57.88±10.23	58.32±9.58	68.25±8.67	68.15±7.98
NMFAD	96.99±1.01	96.90±1.19	83.88±0.65	82.42±1.90	99.04±0.10	98.99±0.17	94.15±1.84	93.72±2.27

contrast, NMFAD explores the strategies of mask-filling and distribution smoothing to maximize the reconstruction errors of anomalies, which do enhance the detection performance.

F. Ablation Experiment

We design two variants for the ablation experiment to highlight the usefulness of NMFAD's motivation:

- 1) **wo-pd**: It is NMFAD without mask-filling and distribution smoothing, which is the basic framework with mask strategy.
- 2) **wo-d**: It is NMFAD without distribution smoothing to verify the usefulness of mask-filling.

The results are shown in Table IV. We make the following observations.

- 1) On each dataset, **wo-pd** is the weakest of the three models, even though it uses the mask strategy to enhance the model's learning ability.
- 2) The superiority of mask-filling can be illustrated by the fact that **wo-d** outperforms **wo-pd** on all datasets.
- 3) After adding the distribution smoothing to **wo-d**, NMFAD performs the best results on all datasets, which could verify that the combination of mask-filling and distribution smoothing presents the largest improvement.

In summary, graph anomaly detection protocols by maximizing reconstruction error of anomalies through mask-adding and smoothing the abnormal nodes closer to the distribution of normal nodes could significantly improve the detection performance of NMFAD in attributed networks.

G. Distribution Smoothing Experiment

Distribution smoothing is an important operation of NMFAD, and we choose KL divergence to implement it. In this experiment, we chose another three manners for distribution smoothing, Wasserstein, Hellinger, Bhattacharyya

TABLE IV
ABLATION RESULTS OF NMFAD

	AUC	AP	AUC	AP
	Cora		Citeseer	
wo-pd	86.11 ± 1.23	84.96 ± 3.41	72.12 ± 2.20	70.26 ± 1.63
wo-d	89.06 ± 2.79	88.91 ± 4.08	75.42 ± 1.38	74.35 ± 1.67
NMFAD	96.99 ± 1.01	96.90 ± 1.19	83.88 ± 0.65	82.42 ± 1.90
	Flickr		BlogCatalog	
Wo-pd	94.95 ± 0.90	94.21 ± 1.10	83.07 ± 4.59	79.58 ± 3.90
Wo-d	97.60 ± 0.49	97.22 ± 0.93	88.62 ± 0.59	85.72 ± 1.71
NMFAD	99.04 ± 0.10	98.99 ± 0.17	94.15 ± 1.84	93.72 ± 2.27

distances, to compare with KL divergence, and the experimental results are shown in Table V.

- 1) Combined with the results in Table III and Table IV, no matter what manner is adopted, the distribution smoothing strategy plays a positive role, which could verify that distribution smoothing is the necessary operation of NMFAD for increasing the reconstruction difficulty of anomalies.
- 2) KL divergence achieves the best results on each dataset, which is why we choose it for distribution smoothing.

H. Computational Experiment

To further illustrate the practicality of NMFAD, we analyze the efficiency of NMFAD and some baselines by the number of parameters, FLOPs (Floating-point Operations) and test time (s) on Flickr dataset. The results are shown in Tab. VI. Among

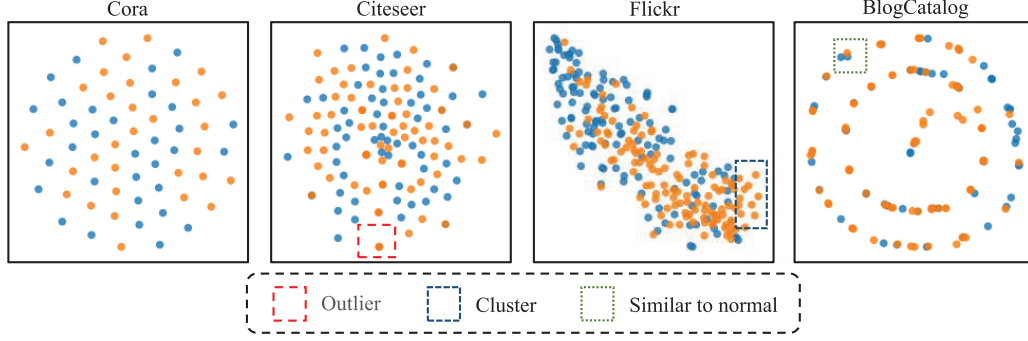


Fig. 6. The normal/abnormal node embedding distribution with mask-filling of NMFAD.

 TABLE V
DISTRIBUTION SMOOTHING RESULTS

	AUC	AP	AUC	AP
	Cora		Citeseer	
Wasserstein	95.59±1.44	95.78±1.16	81.55±1.32	81.30±1.50
Hellinger	95.84±1.46	96.06±1.38	83.49±1.08	82.09±2.26
Bhattacharyya	96.04±1.81	96.21±1.57	83.49±1.11	82.11±2.27
KL divergence	96.99±1.01	96.90±1.19	83.88±0.65	82.42±1.90
	Flickr		BlogCatalog	
Wasserstein	98.92±0.29	98.87±0.35	90.22±1.73	87.36±2.90
Hellinger	98.98±0.06	98.92±0.1	89.40±1.24	85.84±2.75
Bhattacharyya	98.96±0.19	98.89±0.31	88.72±1.95	86.20±2.29
KL divergence	99.04±0.10	98.99±0.17	94.15±1.84	93.72±2.27

 TABLE VI
COMPUTATIONAL RESULTS ON FLICKR DATASET

Methods	Params/MB	FLOPs/G	Test time/s
Deep-SVDD(Attr)	1.5669	11.8669	0.001
Deep-SVDD(Stru)	1.5666	0.0024	0.005
GAE	4.6834	0.1878	0.016
Dominant	4.6834	0.1878	0.092
OC-GCN	1.5666	0.0024	0.005
OC-SAGE	148.3085	1123.3434	0.135
Dual-SVDAE	3.1458	23.7339	0.023
CCA-SSG	1.566592	0.004848	8.522
GraphMAE	4.3238	22.4693	0.012
ComGA	32.4407	245.6421	0.119
NMFAD	6.4433	48.8058	0.030

all the methods, these computational results of NMFAD are in the middle of the range, because it uses a lightweight GCN as the encoder and simple mask-filling operation rather than a more complex network or process, such as OC-SAGE and ComGA. Combined with the results in the “Comparative experiment”, NMFAD can effectively boost the detection capability on the basis of practicability requirements.

I. Visualization Experiments

We conduct two visualization experiments, one is to show the distribution of mask-filled anomalies; one is to summarize the anomaly score distribution.

- 1) To verify the motivation of NMFAD that the reconstruction of the masked anomalous nodes after mask-filling is more like normal rather than abnormal nodes, so that the reconstruction error of the abnormal nodes is large enough to be detected, we visualized the embeddings of the test set for each dataset using the T-SNE tool [47]. The results are shown in Fig. 6 (The blue dots represent normal nodes, and the orange dots represent mask-filled abnormal nodes).

As can be seen, the mask-filled abnormal nodes belonging to the same distribution as the normal nodes in the latent space, would be reconstructed more like the normal nodes and calculate a larger anomaly score based on reconstruction error. Therefore, these visualization results could verify that the combination of mask-filling and distribution smoothing can limit the reconstruction of anomalies that are similar to normal, so as to be detected.

Moreover, in Fig. 6, there are some structural anomalies, such as outlier and cluster. These kinds of anomalous nodes could also be smoothed to the normal distribution after mask-filling, and hard to be reconstructed. Therefore, these visualization results also could indicate that NMFAD can handle some structural anomaly detection tasks.

- 2) Furthermore, we compare an anomaly score distribution of Dual-SVDAE, GraphMAE, ComGA, and NMFAD, which are the methods that performed best in the “Comparative experiment” overall. The results are shown in Fig. 7. As can be seen, NMFAD obtains better normal/abnormal anomaly score distribution than the other methods on all datasets. All the results of Fig. 6, Fig. 7, and Table II highlight that NMFAD can effectively capture anomalies, using the reconstruction error between their mask-filled reconstructed features and original features based on smoothing the abnormal nodes closer to the distribution of normal nodes.

V. DISCUSSION

The above experiments could fully demonstrate that the proposed attributed network detection strategy can effectively improve the detection performance, making anomalies mask-filled by neighbors difficult to be reconstructed, provided that normal nodes can still be reconstructed well.

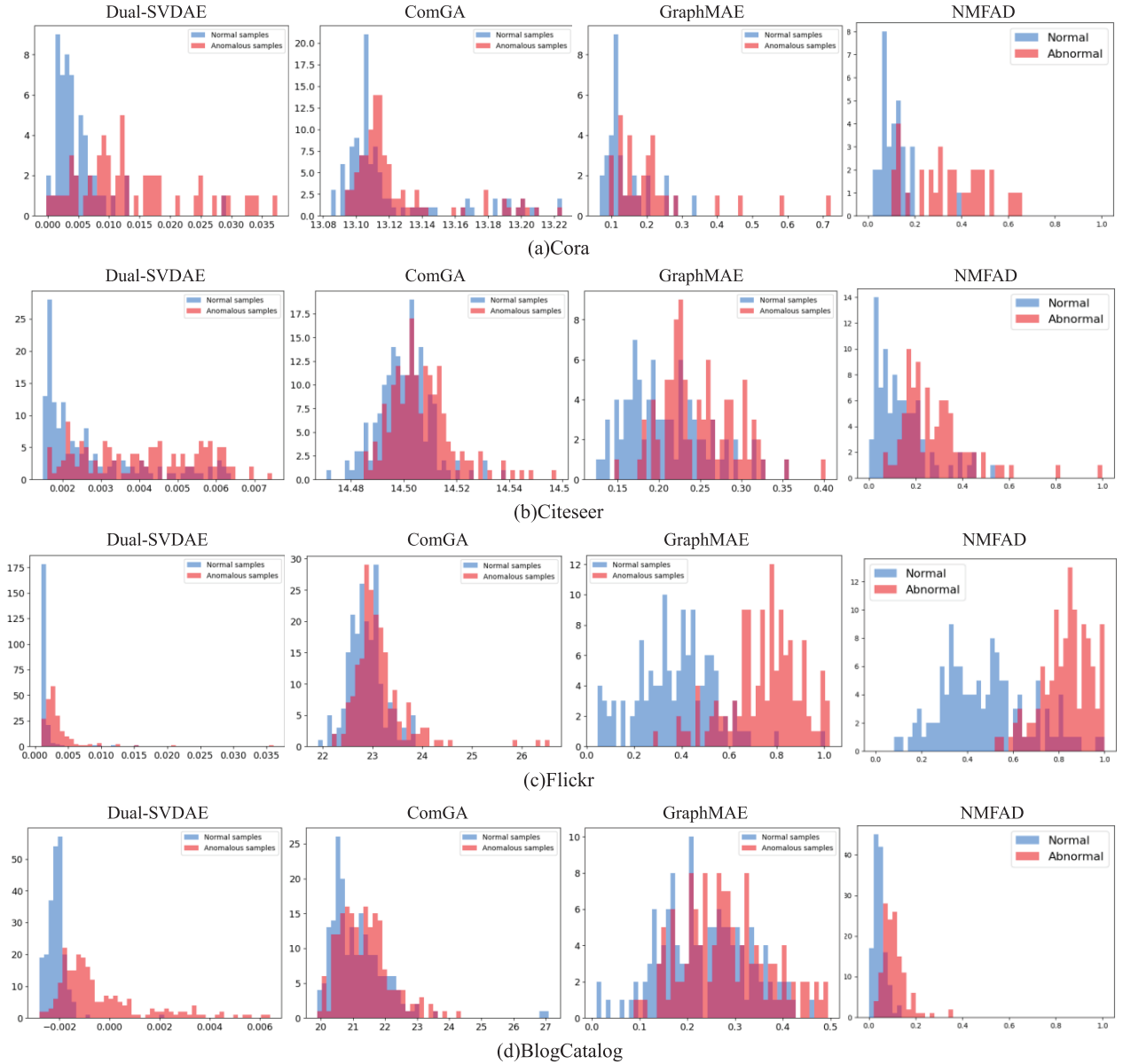


Fig. 7. The anomaly score distribution of Dual-SVDAE, GraphMAE, ComGA, and NMFAD.

Moreover, we employ a lightweight GCN to offset some computational cost of the mask-filling and distribution smoothing operations. The **Computational experiments** could also suggest that NMFAD can be applied to many real-time scenarios.

However, we do not apply NMFAD to large graph or dynamic graph scenarios, mainly due to the specific characteristics of these scenarios, such as highly computation, temporal-evolving graph structure, concept drift, etc, which are required for targeted design. Also, this paper lacks validation of NMFAD in real-world scenarios. All these works are what we want to further study.

VI. CONCLUSION

Attributed network is a typical kind of graph data. Most existing graph anomaly detection methods are mainly based on feature extraction to solely enhance the models'

learning ability, without targeted design for anomaly detection tasks, ignoring the strong/weak similarities between normal/abnormal nodes and neighbors, making them incompetent to capture anomalies in attributed networks. Therefore, we fully consider the dramatically different effects of filling on normal/abnormal samples by neighbors' information, and propose a novel Neighbor-aware Masked Attributed Network Anomaly Detection method (NMFAD), whose ideas of mask, filling, and distribution smoothing could transform abnormal nodes closer to normal ones and maximize the reconstruction error of anomalies to capture all anomalies as much as possible. The experimental results show that NMFAD outperforms the baseline schemes on four datasets and highlight the effectiveness and advantages of this novel graph anomaly detection approach.

However, we did not test NMFAD in large graph or dynamic graph scenarios, which are the focus of future work. Moreover,

we will continue to explore the associations between nodes and neighbors, further update and examine it in some real-time attributed network anomaly detection scenarios, such as cybersecurity, financial fraud detection, industrial production monitoring, etc.

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